

Q4 gennaio 2020

$$X_1, X_2, X_3 \sim U(0, 1) \quad \text{INDIP.}$$

$$M = \max(X_1, X_2, X_3) \quad F_M(a)?, f_M(a)?, E[M]$$

$$\begin{aligned} F_M(a) &= P(M \leq a) = P(\max(X_1, X_2, X_3) \leq a) = \\ &= P(X_1 \leq a, X_2 \leq a, X_3 \leq a) \stackrel{\text{INDIP.}}{=} P(X_1 \leq a) P(X_2 \leq a) P(X_3 \leq a) = \\ &= F_{X_1}(a) F_{X_2}(a) F_{X_3}(a) \end{aligned}$$

$$F_{X_k}(a) = P(X_k \leq a) = \begin{cases} 0 & \text{se } a < 0 \\ a & \text{se } a \in [0, 1] \\ 1 & \text{se } a > 1 \end{cases}$$

$$- \quad \quad \quad | \quad 0 \quad \text{se } a < 0$$

$$F_H(a) = \begin{cases} 0 & \& a < 0 \\ a^3 & \& a \in [0, 1] \\ 1 & \& a > 1 \end{cases}$$

$$f_H(a) = \frac{dF_H}{da} = \begin{cases} 3a^2 & \& a \in [0, 1] \\ 0 & \& \text{otherwise} \end{cases}$$

$$E[H] = \int_{-\infty}^{+\infty} a f_H(a) da = \int_0^1 a \cdot 3a^2 da = \left[3 \frac{a^4}{4} \right]_0^1 = \frac{3}{4}$$

$$T = \min(X_1, X_2, X_3)$$

$$\begin{aligned} F_T(a) &= P(T \leq a) = P(\min(X_1, X_2, X_3) \leq a) = \\ &= 1 - P(\min(X_1, X_2, X_3) > a) = \\ &= 1 - P(X_1 > a, X_2 > a, X_3 > a) = \end{aligned}$$

$$\begin{aligned}
&= 1 - P(X_1 > a, X_2 > a, X_3 > a) \stackrel{F_1, \text{ i.i.d. } P.}{=} \\
&= 1 - P(X_1 > a) P(X_2 > a) P(X_3 > a) = \\
&= 1 - (1 - F_{X_1}(a)) (1 - F_{X_2}(a)) (1 - F_{X_3}(a))
\end{aligned}$$

$$1 - F_{X_k}(a) = \begin{cases} 1 & \text{se } a < 0 \\ 1 - a & \text{se } a \in [0, 1] \\ 0 & \text{se } a > 1 \end{cases}$$

$$\begin{aligned}
F_T(a) &= 1 - (1 - F_{X_1}(a)) (1 - F_{X_2}(a)) (1 - F_{X_3}(a)) = \\
&= 1 - \begin{cases} 1 & \text{se } a < 0 \\ (1 - a)^3 & \text{se } a \in [0, 1] \\ 0 & \text{se } a > 1 \end{cases} =
\end{aligned}$$

$$= \begin{cases} 0 & \text{se } a < 0 \\ 1 - (1-a)^3 & \text{se } a \in [0, 1] \\ 1 & \text{se } a > 1 \end{cases}$$

$$f_T(a) = \frac{dF_T(a)}{da} = \begin{cases} 3(1-a)^2 & \text{se } a \in [0, 1] \\ 0 & \text{altrove} \end{cases}$$

$$E[T] = \int_{-\infty}^{+\infty} a f_T(a) da = \int_0^1 a 3(1-a)^2 da =$$

$$= 3 \int_0^1 a (1 + a^2 - 2a) da = 3 \left[\frac{a^2}{2} + \frac{a^4}{4} - \frac{2a^3}{3} \right]_0^1 =$$

$$= 3 \left(\frac{1}{2} + \frac{1}{4} - \frac{2}{3} \right) = 3 \frac{6 + 3 - 8}{12} = \frac{1}{4}$$

es.

(X, Y) coppia di v.c. congiuntamente continue

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$$f(x, y) = \begin{cases} c(x+y) & \text{se } x, y \in [0, 1] \\ 0 & \text{altrove} \end{cases}$$

c? $f_X(x)$? $f_Y(y)$? X e Y sono indep?

$S = \max(X, Y)$ $F_S(a)$?

$Z = X + Y$ $F_Z(a)$?

c? $f(x, y) \geq 0$ $c \geq 0$

$$1 = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} f(x, y) dx dy = \int_0^1 \int_0^1 c(x+y) dx dy =$$

$$= c \int_0^1 \left(\int_0^1 (x+y) dx \right) dy = c \int_0^1 \left[\frac{x^2}{2} + xy \right]_0^1 dy =$$

$$= C \int_0^1 \left(\int_0^1 (x+y) dx \right) dy = C \int_0^1 \left[\frac{x^2}{2} + xy \right]_0^1 dy =$$

$$= C \int_0^1 \left(\frac{1}{2} + y \right) dy = C \left[\frac{1}{2}y + \frac{y^2}{2} \right]_0^1 = C \left(\frac{1}{2} + \frac{1}{2} \right) = C$$

$$\Rightarrow C = 1$$

$$f(x,y) = \begin{cases} x+y & \text{se } x,y \in [0,1] \\ 0 & \text{altrimenti} \end{cases}$$

$$f_X(x) = \int_{-\infty}^{+\infty} f(x,y) dy =$$

$$\Rightarrow \int_0^1 f(x,y) dy = \begin{cases} \int_0^1 (x+y) dy & \text{se } x \in [0,1] \\ 0 & \text{altrimenti} \end{cases} =$$

se $x \notin [0,1]$
 $f(x,y) = 0$

$$= \int_0^1 \left[xy + \frac{y^2}{2} \right]_0^1 = x + \frac{1}{2} \quad \text{se } x \in [0,1]$$

altrimenti

- 1 0

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Per simmetria si trova facilmente che

$$f_Y(y) = \int_{-\infty}^{+\infty} f(x,y) dx = \begin{cases} y + \frac{1}{2} & \text{se } y \in [0,1] \\ 0 & \text{altrove} \end{cases}$$

$$f(x,y) \neq f_X(x) f_Y(y) \quad \text{No}$$

$\Rightarrow X$ e Y SONO DIPENDENTI.

$$S = \max(X, Y)$$

$$\begin{aligned} F_S(a) &= P(S \leq a) = P(\max(X, Y) \leq a) = \\ &= P(X \leq a, Y \leq a) = \int_{-\infty}^a \int_{-\infty}^a f(x,y) dx dy = \end{aligned}$$

$$= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (x+y) dx dy$$

$$= \begin{cases} 0 & \text{se } a < 0 \\ \int_0^a \int_0^a (x+y) dx dy & \text{se } a \in [0, 1] \\ 1 & \text{se } a > 1 \end{cases}$$

$$= \begin{cases} 0 & \text{se } a < 0 \\ a^3 & \text{se } a \in [0, 1] \\ 1 & \text{se } a > 1 \end{cases}$$

$$(*) = \int_0^a \left(\int_0^a x+y dx \right) dy = \int_0^a \left[\frac{x^2}{2} + xy \right]_0^a dy =$$

$$= \int_0^a \left(\frac{a^2}{2} + ay \right) dy = \left[\frac{a^2}{2} y + \frac{ay^2}{2} \right]_0^a = a^3$$

$$Z = X + Y$$

$$F_Z(a) = P(Z \leq a) = P(X + Y \leq a) = P(Y \leq a - X) =$$

$$= \int_{-\infty}^{+\infty} \left(\int_{-\infty}^{a-x} f(x, y) dy \right) dx = \begin{cases} 0 & \text{se } a < 0 \\ ? & \\ 1 & \text{se } a > 2 \end{cases}$$

$$\text{se } a \in [0, 2]$$

$$0 \leq a - x \leq 1$$

$$-a \leq -x \leq 1 - a$$

$$a - 1 \leq x \leq a$$

$$\begin{aligned} & \int_{-\infty}^{+\infty} \left(\int_0^{a-x} f(x, y) dy \right) dx = \\ & = \begin{cases} \int_{a-1}^a \left(\int_0^{a-x} x+y dy \right) dx & \text{se } a \in [1, 2] \\ \int_0^a \left(\int_0^{a-x} x+y dy \right) dx & \text{se } a \in [0, 1] \end{cases} \end{aligned}$$

Q6 gennaio 2020

$$\bar{X} = 7.5$$

$$S^2 = 16$$

$$n = 51$$

popolazione gaussiana $N(\mu, \sigma)$

$$\mu \in \left[\bar{X} - \frac{S}{\sqrt{n}} t_{n-1, \frac{\alpha}{2}}, \bar{X} + \frac{S}{\sqrt{n}} t_{n-1, \frac{\alpha}{2}} \right]$$

$$1 - \alpha = 95\% \Rightarrow \alpha = 5\% \Rightarrow \frac{\alpha}{2} = 2.5\% = 0.025$$

$$S = \sqrt{S^2} = 4, \quad \sqrt{n} = \sqrt{51}, \quad t_{50, 0.025} \approx 2$$

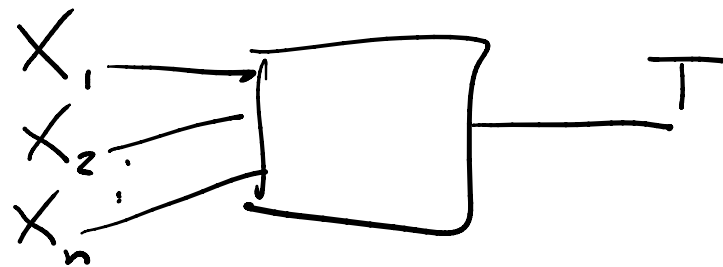
$$\begin{aligned} \mu \in \left[7.5 - \frac{4}{\sqrt{51}} \cdot 2, 7.5 + \frac{4}{\sqrt{51}} \cdot 2 \right] &= [6.379 \dots, 8.620] \\ &= [6.3, 8.6] \end{aligned}$$

$$\sigma^2 \in \left[\frac{(n-1)S^2}{\chi^2_{n-1, \frac{\alpha}{2}}}, \frac{(n-1)S^2}{\chi^2_{n-1, 1-\frac{\alpha}{2}}} \right] = \left[\frac{50 \times 16}{71.42}, \frac{50 \times 16}{32.36} \right] =$$

$$\chi^2_{n-1, \frac{\alpha}{2}} = \chi^2_{50, 0.025} = 71.42 \quad \left| \quad = [11.20, 24.72] \right.$$

$$\chi^2_{n-1, 1-\frac{\alpha}{2}} = \chi^2_{50, 0.975} = 32.36 \quad \left| \quad = [11.2, 24.7] \right.$$

REGRESSIONE



$$T = T(X_1, X_2, \dots, X_n)$$

$$\text{Se } T = \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_n X_n + \alpha$$

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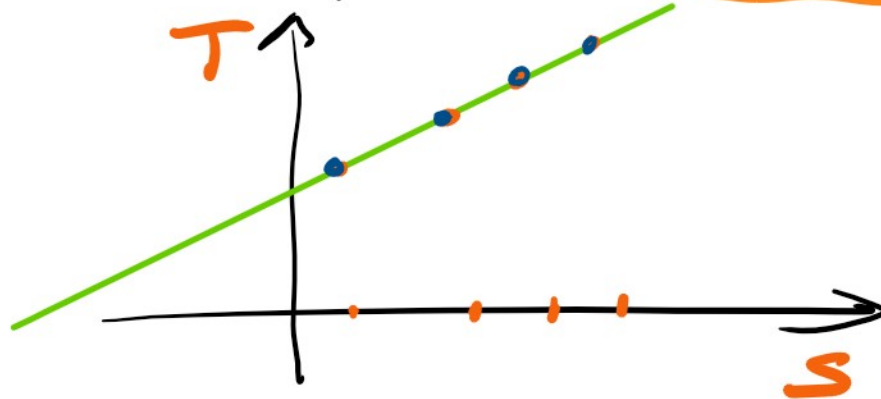
regressione lineare

Se $n=1$ $T=T(X_1)$ regressione semplice

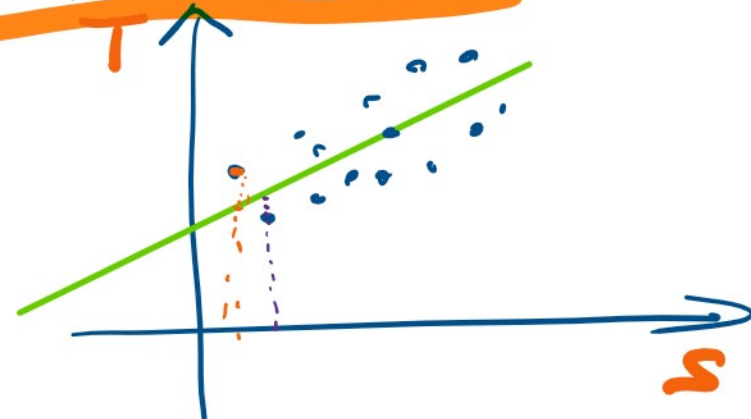
⇒ Regressione lineare semplice

$$T = \beta S + \alpha$$

Nei casi pratici: $T = \beta S + \alpha + \text{errore casuale}$



In assenza
di errore casuale
correlazione immediata



In presenza
di errore casuale
i dati non sono allineati.

di errore casuale
sarebbe immediato
calcolare β e α

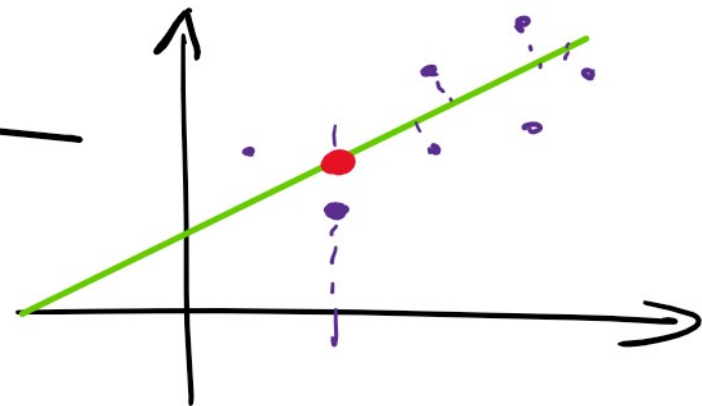
di errore casuale
i dati non sono allineati:

In presenza di errori casuali è possibile
identificare β e α in maniera da rendere la retta
il più vicino possibile ai punti:

$$T = BS + A \quad \text{con } A \text{ e } B \text{ stimatori di } \alpha \text{ e } \beta$$

Dagli esperimenti sono note le misure (S_k, T_k)

S	S_1	S_2	...
T	T_1	T_2	



$$SS = \sum_{k=1}^n (BS_k + A - T_k)^2$$

metodo dei minimi quadrati

metodo dei minimi quadrati:
(chiediamo che la somma dei quadrati SS sia minima)

$$\begin{cases} \frac{\partial SS}{\partial A} = 0 \\ \frac{\partial SS}{\partial B} = 0 \end{cases} \Rightarrow \begin{cases} 2 \sum_{k=1}^n (BS_k + A - T_k) = 0 \\ 2 \sum_{k=1}^n S_k (BS_k + A - T_k) = 0 \end{cases} \quad \text{⊗}$$

$$\bar{S} = \sum_{k=1}^n \frac{S_k}{n}; \quad \bar{T} = \sum_{k=1}^n \frac{T_k}{n} \quad \text{medie aritmetiche NON campionarie}$$

$$\begin{cases} \text{⊗} \Rightarrow B \sum_{k=1}^n S_k + A \sum_{k=1}^n 1 - \sum_{k=1}^n T_k = 0 \\ B \sum_{k=1}^n S_k^2 + A \sum_{k=1}^n S_k - \sum_{k=1}^n S_k T_k = 0 \end{cases} \Rightarrow$$

$$\Rightarrow \begin{cases} B n \bar{S} + A n - n \bar{T} = 0 \\ B \sum_{k=1}^n S_k^2 + A n \bar{S} - \sum_{k=1}^n S_k T_k = 0 \end{cases} \Rightarrow$$

$$\Rightarrow \begin{cases} A = \bar{T} - B \bar{S} \\ B \sum_{k=1}^n S_k^2 + n \bar{S} \bar{T} - B n \bar{S}^2 - \sum_{k=1}^n S_k T_k = 0 \end{cases}$$

$$B = \frac{\sum_{k=1}^n S_k T_k - n \bar{S} \bar{T}}{\sum_{k=1}^n S_k^2 - n \bar{S}^2} = \frac{\sum_{k=1}^n T_k (S_k - \bar{S})}{\sum_{k=1}^n S_k^2 - n \bar{S}^2}$$

$\uparrow$
 $n \bar{T} = \sum_{k=1}^n T_k$

IPOTESI

1) $S_1, S_2, \dots, S_n$ valori di S assegnati in ingresso
 siano noti senza errore (errore di misura
 trascurabile) $\Rightarrow S_k$ sono dei numeri e non

trascurabile) $\Rightarrow S_k$ sono dei numeri e non delle variabili casuali

2) $T_1, \dots, T_n$ sono invece v.c. (errore non trascurabile) e

$$T_k \sim N(\beta S_{k+1}, \sigma^2)$$

$$\begin{aligned} T_k &= \beta S_{k+1} + \varepsilon \quad \text{con } \varepsilon \sim N(0, \sigma^2) \\ T_k &\sim N(\beta S_{k+1}, \sigma^2) \end{aligned}$$

in altre parole immaginiamo che

$$T_k = \beta S_{k+1} + \text{errore} \quad \text{con errore} \sim N(0, \sigma^2)$$

3) $T_1, \dots, T_n$ sono INDIPENDENTI

$$B = \frac{\sum_{k=1}^n (S_k - \bar{S}) T_k}{\sum_{k=1}^n S_k^2 - n \bar{S}^2}$$

$$\begin{aligned} A &= \bar{T} - B \bar{S} = \\ &= \sum_{k=1}^n \frac{T_k}{n} - B \sum_{k=1}^n \frac{S_k}{n} \end{aligned}$$

~ ~ ~

$$\begin{aligned}
 E[B] &= E \left[\frac{\sum_{k=1}^n (S_k - \bar{S}) T_k}{\sum_{k=1}^n S_k^2 - n \bar{S}^2} \right] = \frac{\sum_{k=1}^n (S_k - \bar{S}) E[T_k]}{\sum_{k=1}^n S_k^2 - n \bar{S}^2} = \\
 &= \frac{\sum_{k=1}^n (S_k - \bar{S}) (\beta S_k + \alpha)}{\sum_{k=1}^n S_k^2 - n \bar{S}^2} = \frac{\beta \left(\sum_{k=1}^n S_k^2 - \bar{S} \sum_{k=1}^n S_k \right) + \alpha \left(\sum_{k=1}^n (S_k - \bar{S}) \right)}{\sum_{k=1}^n S_k^2 - n \bar{S}^2} \\
 &= \frac{\beta \left(\sum_{k=1}^n S_k^2 - n \bar{S}^2 \right) + \alpha \left(\cancel{\left(\sum_{k=1}^n S_k \right) - n \bar{S}} \right)}{\sum_{k=1}^n S_k^2 - n \bar{S}^2} = \beta
 \end{aligned}$$

quindi B è uno
stimatore corretto
per β

$$E[A] = E[\bar{T} - B \bar{S}] = E \left[\sum_{k=1}^n T_k - B \bar{S} \right] =$$

$$\begin{aligned}
 E[H] &= E\left[1 - BS\right] = E\left[\frac{\sum_{k=1}^n T_k}{n} - BS\right] = \\
 &= E\left[\frac{\sum_{k=1}^n T_k}{n} - B\bar{S}\right] = \sum_{k=1}^n \frac{E[T_k]}{n} - \bar{S}E[B] = \\
 &= \frac{\sum_{k=1}^n (\beta S_k + d)}{n} - \beta \sum_{k=1}^n \frac{S_k}{n} = \\
 &= \beta \left(\cancel{\frac{\sum_{k=1}^n S_k}{n}} - \cancel{\frac{\sum_{k=1}^n S_k}{n}} \right) + \frac{d}{n} = d
 \end{aligned}$$

A è uno stimatore corretto per d

$$B = \frac{\sum_{k=1}^n (S_k - \bar{S}) T_k}{\sum_{k=1}^n S_k^2 - n\bar{S}^2}$$

$$\sum_{k=1}^n S_k^2 - n\bar{S}^2 = \sum_{k=1}^n S_k^2 - \bar{S} \sum_{k=1}^n S_k = \sum_{k=1}^n S_k (S_k - \bar{S})$$

$$\sum_{k=1}^n S_k - n\bar{S} = \sum_{k=1}^n S_k - S \sum_{k=1}^n 1 = \sum_{k=1}^n (S_k - \bar{S})$$

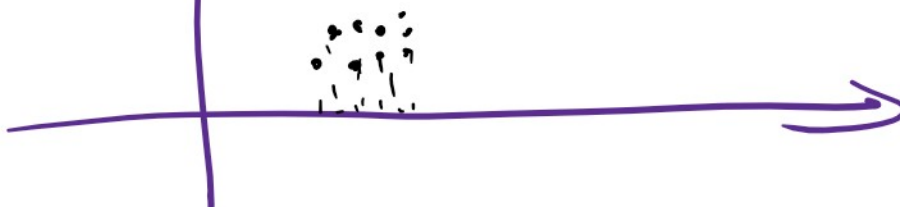
$$\text{Var}(B) = \text{Var} \left(\frac{\sum_{k=1}^n (S_k - \bar{S}) T_k}{\sum_{k=1}^n S_k^2 - n\bar{S}^2} \right) \stackrel{\text{i.i.d.}}{=} \frac{1}{\left(\sum_{k=1}^n S_k^2 - n\bar{S}^2 \right)^2} \sum_{k=1}^n (S_k - \bar{S})^2 \text{Var}(T_k) =$$

$$= \frac{1}{\left(\sum_{k=1}^n S_k^2 - n\bar{S}^2 \right)^2} \sum_{k=1}^n (S_k - \bar{S})^2 \text{Var}(T_k) =$$

$$= \frac{\sum_{k=1}^n (S_k - \bar{S})^2 \sigma^2}{\left(\sum_{k=1}^n S_k^2 - n\bar{S}^2 \right)^2} =$$

$$= \frac{\sigma^2}{\sum_{k=1}^n (S_k - \bar{S})^2}$$

$$\begin{aligned} \sum_{k=1}^n (S_k - \bar{S})^2 &= \\ &= \sum_{k=1}^n (S_k^2 + \bar{S}^2 - 2S_k\bar{S}) = \\ &= \sum_{k=1}^n S_k^2 + n\bar{S}^2 - 2\bar{S} \sum_{k=1}^n S_k = \\ &= \sum_{k=1}^n S_k^2 + n\bar{S}^2 - 2n\bar{S}^2 = \sum_{k=1}^n S_k^2 - n\bar{S}^2 \end{aligned}$$

$$\Rightarrow \text{Var}(B) = \frac{\sigma^2}{\sum_{k=1}^n (S_k - \bar{S})^2}$$


$$= \sum_{k=1}^n S_k + n\bar{S} - 2n\bar{S} = \sum_{k=1}^n S_k - n\bar{S}$$

Dato che $B = \frac{\sum_{k=1}^n (S_k - \bar{S}) T_k}{\sum_{k=1}^n S_k^2 - n\bar{S}^2} \sim \mathcal{N}\left(\beta, \frac{\sigma^2}{\sum_{k=1}^n (S_k - \bar{S})^2}\right)$